

Date: Sept 29, 2025

Math Trimester 1

MorningWork

Name:

Class of:

Teacher:

Homerroom:

1) Which expression is equivalent to $100x^2 - 16$?

- (1) $(50x - 8)(50x + 8)$
- (2) $(50x - 8)(50x - 8)$
- (3) $(10x - 4)(10x + 4)$
- (4) $(10x - 4)(10x - 4)$

Solution: $100x^2 - 16 = (10x)^2 - 4^2$ is a difference of squares $\Rightarrow (10x - 4)(10x + 4)$. Choice (3)

3) If $g(x) = -2x^2 + 16$, then $g(-3)$ equals

- (1) -20
- (2) -2
- (3) 34
- (4) 52

Solution: $g(-3) = -2(-3)^2 + 16 = -2(9) + 16 = -18 + 16 = -2$. Choice (2)

4) What are the zeros of $f(x) = x^2 - 8x - 20$?

- (1) 10 and 2
- (2) 10 and -2
- (3) -10 and 2
- (4) -10 and -2

Solution: Solve $x^2 - 8x - 20 = 0$. Factor: $(x - 10)(x + 2) = 0$. Zeros: $x = 10$ and $x = -2$. Choice (2)

5) Which point lies on the graph of $y = 3x^2 - \frac{1}{4}x + 3$?

- (1) $(-2, 15.5)$
- (2) $(-1, 5.75)$
- (3) $(1, 6.25)$
- (4) $(2, 15.5)$

Solution: For $x = -2$: $y = 3(4) - \frac{1}{4}(-2) + 3 = 12 + 0.5 + 3 = 15.5 \Rightarrow (-2, 15.5)$ lies on the graph. (Checks of other choices do not match.) Choice (1)

6) Given $f(x) = x^2$ and $g(x) = 8x - 15$ graphed on the same set of axes, which value(s) of x will make $f(x) = g(x)$?

- (1) 3, only
- (2) 9, only
- (3) 3 and 5
- (4) 9 and 25

Solution: Set $x^2 = 8x - 15 \Rightarrow x^2 - 8x + 15 = 0 \Rightarrow (x - 3)(x - 5) = 0$. Thus $x = 3$ or $x = 5$. Choice (3)

7) Which trinomial is written in standard form and has a constant term of five?

- (1) $x^5 - 4x^2 + 10$
- (2) $2x^2 + 6x^4 + 5$
- (3) $5x^4 - 3x^2 + 1$
- (4) $4x^5 - 8x^2 + 5$

Solution: Standard form orders terms by descending powers. Only $4x^5 - 8x^2 + 5$ is in standard form and has constant 5. Choice (4)